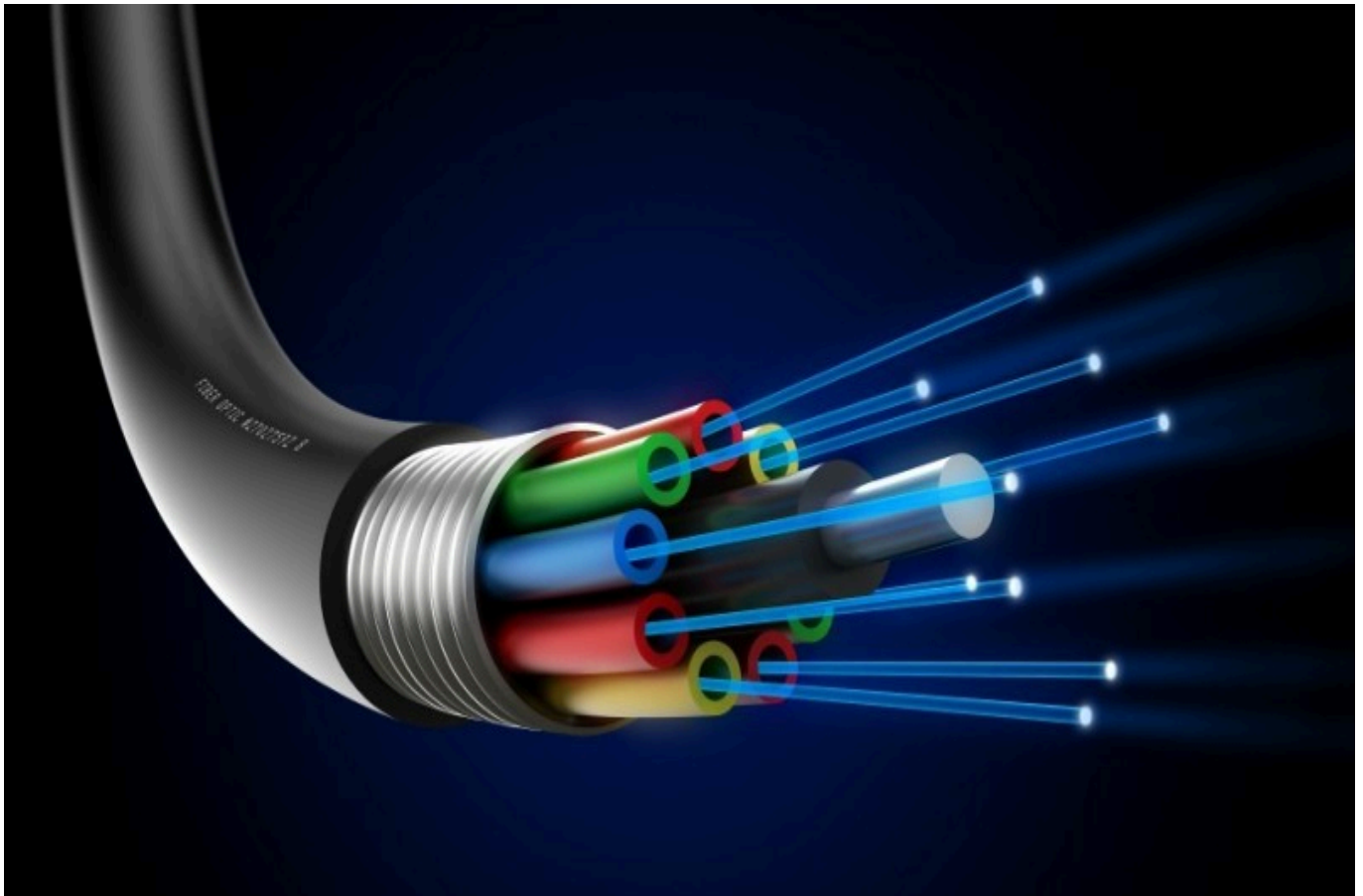


Qualification Pack



Fiber to-the Home (FTTH/X) Installer

QP Code: TEL/Q4200

Version: 4.0

NSQF Level: 3

Telecom || 3rd Floor, Plot No 126, Sector - 44
Gurgaon - 122003 || email:standards@tsscindia.com

Qualification Pack

Contents

TEL/Q4200: Fiber to-the Home (FTTH/X) Installer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
TEL/N4143: Install Fiber-to-the-Home (FTTH/X) and Fiber-to-Anywhere (FTTx) Cables	5
TEL/N6400: Splice Optical Fiber	13
TEL/N4200: Installation of Passive FTTH/X components	22
TEL/N4201: In-building FTTH/X cabling	28
TEL/N4131: Work Safety Practices with Fiber Optics	37
TEL/N9111: Follow sustainability practices in telecom cabling operations	43
DGT/VSQ/N0101: Employability Skills (30 Hours)	50
Assessment Guidelines and Weightage	55
<i>Assessment Guidelines</i>	55
<i>Assessment Weightage</i>	56
Acronyms	57
Glossary	59

Qualification Pack

TEL/Q4200: Fiber to-the Home (FTTH/X) Installer

Brief Job Description

A Fiber to-the Home (FTTH/X) Installer is responsible for implementing fiber optic cabling from the Optical Line Test Set (OLTS) drop point to the customer's termination point, ensuring seamless last-mile connectivity. The job includes fiber splicing, termination at distribution points, and adherence to structured cabling standards, telecommunication regulations, and network safety norms.

Personal Attributes

The role requires strong problem-solving skills, attention to detail, compliance with work instructions, and effective communication to collaborate with team members and stakeholders. Adaptability to emerging technologies and commitment to safety practices are essential.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N4143: Install Fiber-to-the-Home \(FTTH/X\) and Fiber-to-Anywhere \(FTTx\) Cables](#)
2. [TEL/N6400: Splice Optical Fiber](#)
3. [TEL/N4200: Installation of Passive FTTH/X components](#)
4. [TEL/N4201: In-building FTTH/X cabling](#)
5. [TEL/N4131: Work Safety Practices with Fiber Optics](#)
6. [TEL/N9111: Follow sustainability practices in telecom cabling operations](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
Country	India
NSQF Level	3

Qualification Pack

Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.0803
Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass with 3 Years of experience in the operations and maintenance of passive infrastructure OR Previous relevant Qualification of NSQF Level 2.5 with 1.5 years of experience in operations and maintenance of passive infrastructure
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	4.0
Reference code on NQR	QG-03-TL-04083-2025-V2-TSSC
NQR Version	2

Qualification Pack

TEL/N4143: Install Fiber-to-the-Home (FTTH/X) and Fiber-to-Anywhere (FTTx) Cables

Description

This unit is about outside plant installation procedures and practices for FTTH/X and FTTx deployment, covering advancements in optical fiber cable installation techniques, network design considerations, and compliance with industry standards.

Scope

The scope covers the following :

- Pre-installation checks and processes
- Direct buried installation
- Underground (duct) installation
- Aerial installation
- FTTx solutions
- Network planning and design considerations

Elements and Performance Criteria

Pre-installation checks and processes

To be competent, the user/individual on the job must be able to:

- PC1.** conduct pre-deployment surveys, including advanced route mapping using Geographic Information Systems (GIS) tools and FTTx architecture requirements
- PC2.** perform pre-tests of optical cables using OTDR and AI-enabled diagnostic tools to ensure network readiness
- PC3.** ensure cable compliance with advanced standards, including bend-insensitive and high-density cables suitable for FTTx variations
- PC4.** Verify critical parameters such as maximum pulling tension, bend radius, and cable diameter to ensure installation adherence to design specifications
- PC5.** Use duct preparation tools, including duct integrity testing and cleaning with compressed air or vacuum systems, ensuring readiness for cable installation
- PC6.** Ensure necessary permissions and compliance with regulatory requirements, including local laws and environmental guidelines

Direct buried installation

To be competent, the user/individual on the job must be able to:

- PC7.** select appropriate cable types (e.g., armored, non-armored, water-blocking cables) based on site-specific requirements and environmental factors
- PC8.** perform trenching using precision-guided trenching tools, ensuring minimal environmental impact
- PC9.** follow advanced practices for armor bonding and grounding to maintain signal integrity
- PC10.** ensure proper cable laying, covering, and placement of physical markers, adhering to environmental and safety standards

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PC11. perform trench reinstatements using eco-friendly materials to meet sustainability guidelines

Underground (duct) installation

To be competent, the user/individual on the job must be able to:

- PC12.** use automated duct cable pulling tools and accessories like motorized winches and smart pulling grips for efficient installations
- PC13.** execute cable blowing using advanced machines equipped with AI monitoring for friction and duct-fill optimization
- PC14.** test duct suitability, coefficient of friction, and other factors affecting cable longevity using specialized tools
- PC15.** perform figure-8 cable coiling techniques to prevent damage during long-haul installations
- PC16.** integrate multiple ducts with proper sealing and alignment, ensuring scalability and durability

Aerial installation

To be competent, the user/individual on the job must be able to:

- PC17.** handle aerial-specific cables like ADSS cables with tension control tools to ensure stability and longevity
- PC18.** survey poles, messenger strands, and aerial infrastructure for damage or potential challenges using drones and smart tools
- PC19.** install messenger strands and aerial cables safely using lift buckets or similar equipment
- PC20.** conduct wind-loading analysis and implement mitigation techniques to enhance network reliability

FTTx solutions

To be competent, the user/individual on the job must be able to:

- PC21.** install fiber for FTTx variations, including Fiber to the Node (FTTN), Fiber to the Curb (FTTC), Fiber to the Building (FTTB) and Fiber-to-the-Desk (FTTD)
- PC22.** deploy and configure pre-connectorized optical cables, micro-trenching, and micro-ducts for high-density urban applications
- PC23.** integrate network components like ONTs, OLTs, splitters, and IoT devices to create efficient smart home and enterprise networks
- PC24.** ensure seamless splicing and terminations for FTTx networks using precision splicing tools

Network planning and design considerations

To be competent, the user/individual on the job must be able to:

- PC25.** apply network planning and design principles, including selection of the right architecture (PON, GPON) for deployment based on site needs
- PC26.** determine infrastructure requirements, such as conduit paths, fiber type, and distribution points, ensuring optimal resource use
- PC27.** ensure alignment with the applicable regulatory and compliance requirements for FTTH network design
- PC28.** select appropriate equipment and materials for scalability and future upgrades

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** the basics regarding the modern applications of fiber optics like - smart cities, 5G, telemedicine, cloud computing, smart homes, edge computing, e-learning, high-frequency trading, Industry 4.0, AR/VR, smart agriculture, and high-bandwidth entertainment, enabling seamless connectivity, automation, and digital transformation through IoT, edge computing, high-speed connectivity, and low-latency communication
- KU2.** the advances in high-capacity networks, AI-driven optimization, sustainable deployment, and fiber network cybersecurity
- KU3.** the principles of FTTx architectures, including FTTN, FTTC, FTTB and FTTD, and their impact on deployment
- KU4.** network planning and design principles, including selection of architecture (PON, GPON), infrastructure and cabling types
- KU5.** use of GIS tools for mapping fiber infrastructure, route planning, and network asset tracking
- KU6.** the key characteristics of fiber cables like bend-insensitive, ribbon fibers, and high-fiber-count cables
- KU7.** the regulatory and compliance considerations for optical fiber installations, including environmental impact and safety standards
- KU8.** different splicing methods like fusion splicing, mass splicing, mechanical splicing, etc.
- KU9.** termination techniques for connectors and pigtails
- KU10.** cable pulling, blowing, and routing methods
- KU11.** the tools and technologies for testing cables, real-time fault detection and predictive maintenance in fiber networks, including AI-enabled OTDR, power meters, and visual fault locators
- KU12.** advanced installation techniques, such as micro-trenching, air-blown fibers, and robotic pulling
- KU13.** the role of automation and smart tools in testing and diagnostics
- KU14.** the methods for integrating ONTs, OLTs, and IoT devices into FTTx networks
- KU15.** the principles of wind-loading analysis and mitigation techniques for aerial installations
- KU16.** the applicable sustainability practices, including eco-friendly, installations, reinstatements and e-waste management
- KU17.** the applicable Industry standards and guidelines issued by TRAI, Broadband Forum, etc.
- KU18.** the escalation protocols for incidents and challenges during installations
- KU19.** the concepts and applications of micro-trenching and air-blown fiber techniques in high-density urban installations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret advanced technical documents, drawings, and regulatory guidelines
- GS2.** communicate effectively with stakeholders, supervisors, and clients about installation progress and challenges
- GS3.** analyze and resolve installation issues using critical thinking and troubleshooting skills
- GS4.** apply data analytics tools to monitor and improve installation efficiency and network performance

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- GS5.** train and mentor junior technicians on advanced FTTx practices and emerging technologies
- GS6.** adapt to new technologies, tools, and methodologies in fiber installation
- GS7.** ensure compliance with safety, health, and environmental regulations during installations
- GS8.** coordinate with vendors and third-party contractors to procure and maintain equipment
- GS9.** manage time and resources effectively using advanced planning tools

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-installation checks and processes</i>	6	12	-	5
PC1. conduct pre-deployment surveys, including advanced route mapping using Geographic Information Systems (GIS) tools and FTTx architecture requirements	1	2	-	0.5
PC2. perform pre-tests of optical cables using OTDR and AI-enabled diagnostic tools to ensure network readiness	1	2	-	1
PC3. ensure cable compliance with advanced standards, including bend-insensitive and high-density cables suitable for FTTx variations	1	2	-	1
PC4. Verify critical parameters such as maximum pulling tension, bend radius, and cable diameter to ensure installation adherence to design specifications	1	2	-	1
PC5. Use duct preparation tools, including duct integrity testing and cleaning with compressed air or vacuum systems, ensuring readiness for cable installation	1	2	-	1
PC6. Ensure necessary permissions and compliance with regulatory requirements, including local laws and environmental guidelines	1	2	-	0.5
<i>Direct buried installation</i>	5	10	-	4
PC7. select appropriate cable types (e.g., armored, non-armored, water-blocking cables) based on site-specific requirements and environmental factors	1	2	-	0.5
PC8. perform trenching using precision-guided trenching tools, ensuring minimal environmental impact	1	2	-	1
PC9. follow advanced practices for armor bonding and grounding to maintain signal integrity	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. ensure proper cable laying, covering, and placement of physical markers, adhering to environmental and safety standards	1	2	-	1
PC11. perform trench reinstatements using eco-friendly materials to meet sustainability guidelines	1	2	-	0.5
<i>Underground (duct) installation</i>	5	10	-	5
PC12. use automated duct cable pulling tools and accessories like motorized winches and smart pulling grips for efficient installations	1	2	-	1
PC13. execute cable blowing using advanced machines equipped with AI monitoring for friction and duct-fill optimization	1	2	-	1
PC14. test duct suitability, coefficient of friction, and other factors affecting cable longevity using specialized tools	1	2	-	1
PC15. perform figure-8 cable coiling techniques to prevent damage during long-haul installations	1	2	-	1
PC16. integrate multiple ducts with proper sealing and alignment, ensuring scalability and durability	1	2	-	1
<i>Aerial installation</i>	3	6	-	2
PC17. handle aerial-specific cables like ADSS cables with tension control tools to ensure stability and longevity	0.5	1	-	0.5
PC18. survey poles, messenger strands, and aerial infrastructure for damage or potential challenges using drones and smart tools	0.5	1	-	0.5
PC19. install messenger strands and aerial cables safely using lift buckets or similar equipment	1	2	-	0.5
PC20. conduct wind-loading analysis and implement mitigation techniques to enhance network reliability	1	2	-	0.5
<i>FTTx solutions</i>	4	6	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. install fiber for FTTx variations, including Fiber to the Node (FTTN), Fiber to the Curb (FTTC), Fiber to the Building (FTTB) and Fiber-to-the-Desk (FTTD)	1	1	-	0.5
PC22. deploy and configure pre-connectorized optical cables, micro-trenching, and micro-ducts for high-density urban applications	1	1	-	0.5
PC23. integrate network components like ONTs, OLTs, splitters, and IoT devices to create efficient smart home and enterprise networks	1	2	-	0.5
PC24. ensure seamless splicing and terminations for FTTx networks using precision splicing tools	1	2	-	0.5
<i>Network planning and design considerations</i>	7	6	-	2
PC25. apply network planning and design principles, including selection of the right architecture (PON, GPON) for deployment based on site needs	1	1	-	0.5
PC26. determine infrastructure requirements, such as conduit paths, fiber type, and distribution points, ensuring optimal resource use	2	1	-	0.5
PC27. ensure alignment with the applicable regulatory and compliance requirements for FTTH network design	2	2	-	0.5
PC28. select appropriate equipment and materials for scalability and future upgrades	2	2	-	0.5
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4143
NOS Name	Install Fiber-to-the-Home (FTTH/X) and Fiber-to-Anywhere (FTTx) Cables
Sector	Telecom
Sub-Sector	
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N6400: Splice Optical Fiber

Description

This unit focuses on preparing and executing efficient optical splicing operations. It integrates advanced fiber testing tools, automation, network troubleshooting, and sustainable practices, emphasizing precision and efficiency in high-speed optical networks.

Scope

The scope covers the following :

- Check the availability of tools, spares, and advanced equipment
- Prepare cables for splicing for new installations
- Maintain and troubleshoot laid Optical Fiber Cables (OFCs)
- Perform advanced splicing operations with automation and precision
- Utilize advanced fiber testing tools and document results
- Perform specialized splicing techniques for micro and nano fibers

Elements and Performance Criteria

Check the availability of tools, spares, and advanced equipment

To be competent, the user/individual on the job must be able to:

- PC1.** check the availability and functionality of advanced optical testing tools such as AI-enabled Optical Time-Domain Reflectometer (OTDR), power meter, Optical Spectrum Analyzer (OSA), Chromatic Dispersion (CD) Analyzer, and Polarization Mode Dispersion (PMD) Analyzer
- PC2.** check for the availability of tools for advanced splicing including automated splicers, robotic arms, precision cleavers, smart cleavers, splice trays, and inspection tools
- PC3.** check for adequate stock of splicing consumables including joint kits, connectors, patch cords, heat shrink sleeves, and environmental sealing materials along with RCC joint chambers, fiber optic enclosures, and other site-specific materials
- PC4.** ensure splicing machines and testing equipment are calibrated, updated with the latest firmware, and meet precision standards
- PC5.** handle inventory and coordinate repair or replacement of faulty tools through defined channels

Prepare cables for splicing for new installations

To be competent, the user/individual on the job must be able to:

- PC6.** locate and identify fibers for splicing as per network designs, connectivity plans and colour-coded sequences using automated mapping tools
- PC7.** check cables for physical and sheath damage using visual and advanced inspection tools
- PC8.** prepare optical fibers by removing jackets and cleaning cores using automated cleaning systems
- PC9.** secure cables and ensure compliance with bend radius and stress limits specified by manufacturers
- PC10.** install and ground joint closures, splitters, and pigtails with weatherproofing measures to minimize environmental effects

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PC11. manage routing of connectorized fibers inside junction boxes and document compliance with network plans

PC12. perform twist splicing for copper or hybrid cables where applicable

Maintain and troubleshoot laid Optical Fiber Cables (OFCs)

To be competent, the user/individual on the job must be able to:

PC13. identify fiber faults using advanced tools such as AI-enabled OTDR and real-time monitoring systems

PC14. use robotic arms, Optical Fiber Identifiers (OFIs), and smart cleavers for precise maintenance in challenging environments

PC15. coordinate with the Network Operations Center (NOC) for outage windows and route updates

PC16. access fault points safely and inspect for microbends, sheath damage, or environmental wear using advanced inspection tools

PC17. clean and prepare all fibers for splicing as per applicable specifications

PC18. replace damaged sections, re-splice fibers, and ensure all connections meet loss and reflection standards

PC19. secure and weatherproof cables following the applicable standards

PC20. document maintenance and troubleshooting activities digitally for analysis and reporting

Perform advanced splicing operations with automation and precision

To be competent, the user/individual on the job must be able to:

PC21. operate advanced fusion splicing machines with automation features to minimize human error

PC22. perform mechanical, fusion, ribbon, crimp, mass and nano fiber splicing for various network applications, including harsh environments

PC23. ensure splice quality using real-time diagnostics integrated into automated splicing machines

PC24. minimize splice loss and optimize alignment using precision cleavers and advanced imaging tools

PC25. test splice quality using real-time monitoring systems integrated into splicers

PC26. seal splices with heat-shrink or cold-shrink closures, ensuring protection against environmental factors

PC27. conduct micro and nano fiber splicing using specialized tools for high-performance, low-loss connections in specialized industries such as medical and military communications

PC28. document splicing details and results digitally for future reference

Utilize advanced fiber testing tools and document results

To be competent, the user/individual on the job must be able to:

PC29. use AI-enabled OTDR for real-time fault detection, predictive maintenance, and accurate loss measurement

PC30. test signal quality and propagation characteristics using advanced tools such as OSAs, CD analyzers, and PMD analyzers

PC31. verify performance KPIs, including attenuation, reflection, and chromatic dispersion, for spliced fibers

PC32. generate automated reports using cloud-based platforms for remote monitoring and compliance tracking

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PC33. maintain proper documentation of test results for quality assurance and network optimization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the structure, types, and materials of optical fibers, including core, cladding, and jacket properties
- KU2.** the types of splicing techniques such as mechanical, fusion, twist (for copper or hybrid cables), heat shrink, crimp, ribbon, and nano splicing, and their applications
- KU3.** use of tools and equipment for splicing including splice trays, fusion splicers, hand tools (strippers, cutters, crimpers), inspection tools (microscopes, OTDR for fiber splicing), splice closures, and safety equipment (gloves, goggles).
- KU4.** advanced characteristics of optical fibers such as dispersion, refraction, attenuation, and signal loss
- KU5.** features and functions of advanced splicing machines, robotic arms, smart cleavers, and testing equipment
- KU6.** best practices for minimizing splice loss and ensuring long-term durability of joints
- KU7.** the techniques for splicing in challenging environments such as outdoor, submarine, or underground networks
- KU8.** fiber pigtails, connectorized fiber, routing of fiber inside the junction box
- KU9.** different fiber jointing techniques
- KU10.** environmental effects on fiber and splice joints and methods for mitigating degradation
- KU11.** the applications of AI-powered tools for fault detection, predictive maintenance, and process optimization
- KU12.** the principles of cloud-based systems for remote monitoring, reporting, and troubleshooting
- KU13.** the integration of splicing tasks with IoT-enabled smart network management systems
- KU14.** the regulatory compliance and sustainability practices for optical fiber installation and maintenance
- KU15.** fiber optic cable construction, including the structure of optical fibers (core, cladding, jacket), types of fibers (single-mode vs. multi-mode), and fiber optic cable preparation
- KU16.** the advanced fusion splicing process, including an introduction to fusion splicing, fiber preparation (strip, clean, cleave), fusion splicing machine operation, and aligning fibers to create a splice
- KU17.** advanced fiber optic splicing - ribbon fiber splicing techniques, splicing in high-density and harsh environments (outdoor and submarine cables), and achieving high-performance low-loss connections
- KU18.** splice closure and protection including the proper use of splice closures, heat shrink vs. cold shrink closures, and weatherproofing and sealing techniques
- KU19.** the best practices for splicing including minimizing splice loss, accounting for environmental factors, and proper documentation and labeling for splicing
- KU20.** the techniques for crimp splicing and its applications in hybrid networks
- KU21.** cold shrink closures and their advantages in specific environments

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and report all splicing and testing activities using digital tools
- GS2.** work collaboratively with team members, supervisors, and NOC teams for efficient network management
- GS3.** prioritize tasks to meet deadlines during installation, maintenance, and splicing activities
- GS4.** troubleshoot fiber faults and tool malfunctions effectively to minimize downtime
- GS5.** ensure compliance with safety protocols and environmental standards during splicing and testing operations
- GS6.** adapt to emerging tools and technologies to enhance splicing precision and efficiency
- GS7.** communicate clearly and foster positive working relationships with stakeholders
- GS8.** implement sustainability practices, including recycling fiber waste and optimizing resource use

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check the availability of tools, spares, and advanced equipment</i>	4	10	-	4
PC1. check the availability and functionality of advanced optical testing tools such as AI-enabled Optical Time-Domain Reflectometer (OTDR), power meter, Optical Spectrum Analyzer (OSA), Chromatic Dispersion (CD) Analyzer, and Polarization Mode Dispersion (PMD) Analyzer	1	2	-	0.5
PC2. check for the availability of tools for advanced splicing including automated splicers, robotic arms, precision cleavers, smart cleavers, splice trays, and inspection tools	1	2	-	0.5
PC3. check for adequate stock of splicing consumables including joint kits, connectors, patch cords, heat shrink sleeves, and environmental sealing materials along with RCC joint chambers, fiber optic enclosures, and other site-specific materials	1	2	-	1
PC4. ensure splicing machines and testing equipment are calibrated, updated with the latest firmware, and meet precision standards	0.5	2	-	1
PC5. handle inventory and coordinate repair or replacement of faulty tools through defined channels	0.5	2	-	1
<i>Prepare cables for splicing for new installations</i>	5	12	-	4
PC6. locate and identify fibers for splicing as per network designs, connectivity plans and colour-coded sequences using automated mapping tools	0.5	2	-	1
PC7. check cables for physical and sheath damage using visual and advanced inspection tools	1	2	-	0.5
PC8. prepare optical fibers by removing jackets and cleaning cores using automated cleaning systems	1	2	-	0.5
PC9. secure cables and ensure compliance with bend radius and stress limits specified by manufacturers	0.5	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. install and ground joint closures, splitters, and pigtails with weatherproofing measures to minimize environmental effects	0.5	2	-	1
PC11. manage routing of connectorized fibers inside junction boxes and document compliance with network plans	0.5	1	-	0.5
PC12. perform twist splicing for copper or hybrid cables where applicable	1	1	-	0.5
<i>Maintain and troubleshoot laid Optical Fiber Cables (OFCs)</i>	8	12	-	4
PC13. identify fiber faults using advanced tools such as AI-enabled OTDR and real-time monitoring systems	1	2	-	0.5
PC14. use robotic arms, Optical Fiber Identifiers (OFIs), and smart cleavers for precise maintenance in challenging environments	1	1	-	0.5
PC15. coordinate with the Network Operations Center (NOC) for outage windows and route updates	1	1	-	0.5
PC16. access fault points safely and inspect for microbends, sheath damage, or environmental wear using advanced inspection tools	1	1	-	0.5
PC17. clean and prepare all fibers for splicing as per applicable specifications	1	2	-	0.5
PC18. replace damaged sections, re-splice fibers, and ensure all connections meet loss and reflection standards	1	2	-	0.5
PC19. secure and weatherproof cables following the applicable standards	1	2	-	0.5
PC20. document maintenance and troubleshooting activities digitally for analysis and reporting	1	1	-	0.5
<i>Perform advanced splicing operations with automation and precision</i>	8	10	-	4
PC21. operate advanced fusion splicing machines with automation features to minimize human error	1	1	-	0.5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. perform mechanical, fusion, ribbon, crimp, mass and nano fiber splicing for various network applications, including harsh environments	1	2	-	0.5
PC23. ensure splice quality using real-time diagnostics integrated into automated splicing machines	1	1	-	0.5
PC24. minimize splice loss and optimize alignment using precision cleavers and advanced imaging tools	1	1	-	0.5
PC25. test splice quality using real-time monitoring systems integrated into splicers	1	2	-	0.5
PC26. seal splices with heat-shrink or cold-shrink closures, ensuring protection against environmental factors	1	1	-	0.5
PC27. conduct micro and nano fiber splicing using specialized tools for high-performance, low-loss connections in specialized industries such as medical and military communications	1	1	-	0.5
PC28. document splicing details and results digitally for future reference	1	1	-	0.5
<i>Utilize advanced fiber testing tools and document results</i>	5	6	-	4
PC29. use AI-enabled OTDR for real-time fault detection, predictive maintenance, and accurate loss measurement	1	2	-	1
PC30. test signal quality and propagation characteristics using advanced tools such as OSAs, CD analyzers, and PMD analyzers	1	1	-	0.5
PC31. verify performance KPIs, including attenuation, reflection, and chromatic dispersion, for spliced fibers	1	1	-	0.5
PC32. generate automated reports using cloud-based platforms for remote monitoring and compliance tracking	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. maintain proper documentation of test results for quality assurance and network optimization	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6400
NOS Name	Splice Optical Fiber
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	3
Credits	5
Version	7.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N4200: Installation of Passive FTTH/X components

Description

This unit covers the installation of passive FTTH/X components, including splitters, ensuring compatibility with GPON and advanced PON technologies like XG-PON and NG-PON2. It includes working with loss budgets, advanced transmission mechanisms (e.g., WDM, TDMA), and conducting advanced testing for optimal network performance

Scope

The scope covers the following :

- Install passive FTTH/X components (Splitters)
- Perform terminal connections (single incoming/multiple outgoing)
- Conduct power and insertion loss tests
- Work with GPON technologies, loss budgets, and advanced transmission mechanisms

Elements and Performance Criteria

Install passive FTTH/X components (Splitters)

To be competent, the user/individual on the job must be able to:

- PC1.** identify and assess passive components, including PLC and FBT splitters, and their deployment environments
- PC2.** install wall-mount and rack-mount splitters (e.g., 1x8, 1x16, 1x32) using precision tools and advanced equipment
- PC3.** ensure compatibility of splitters with GPON and emerging PON technologies like XG-PON and NG-PON2
- PC4.** check installation sites for optimal placement to minimize signal loss, comply with building standards, and address environmental impacts
- PC5.** identify and configure splitters for advanced transmission technologies, including Wavelength Division Multiplexing (WDM) and Time Division Multiple Access (TDMA)

Perform terminal connections (single incoming/multiple outgoing)

To be competent, the user/individual on the job must be able to:

- PC6.** identify and organize feeder and distribution cables/pigtails for efficient routing and minimal signal interference
- PC7.** use advanced connectors (e.g., SC, LC, APC) to establish secure and loss-free connections
- PC8.** apply fiber management techniques to ensure organized, secure, and scalable installations
- PC9.** perform final connector polishing to maximize optical clarity and reduce insertion loss

Conduct power and insertion loss tests

To be competent, the user/individual on the job must be able to:

- PC10.** conduct insertion loss and reflection testing using advanced OLTS and OTDR equipment
- PC11.** measure power output at distribution ports using precision power meters
- PC12.** validate network performance parameters for compliance with GPON and NG-PON2 standards

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PC13. analyze and calculate loss budgets, considering WDM and TDMA technologies, to ensure network design compliance

Work with GPON technologies, loss budgets, and advanced transmission mechanisms

To be competent, the user/individual on the job must be able to:

PC14. install and configure passive components compatible with advanced GPON, XG-PON, and NG-PON2 networks

PC15. identify the role of broadcast technologies in GPON deployments and configure splitters accordingly

PC16. use AI-enabled diagnostic tools for real-time fault detection, insertion loss tracking, and troubleshooting

PC17. analyze the impact of components (e.g., splitters, connectors) on loss budgets and optimize designs using best practices

PC18. validate the configuration and deployment of advanced transmission mechanisms like WDM and TDMA for optimized bandwidth utilization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Passive Optical Networks (PONs) like GPON, EPON, and Next-Gen PON

KU2. GPON technology fundamentals, including its architecture, components, and benefits

KU3. key components of GPON systems, such as ONTs, OLTs, and splitters, and their roles in network deployments

KU4. Outside Plant (OSP) considerations, including routing, environmental factors, and secure installation practices

KU5. advanced transmission mechanisms like WDM and TDMA, their roles, and deployment considerations

KU6. loss budget concepts, calculation methods, the impact of components on loss budget and best design practices for minimizing optical losses

KU7. power testing techniques, including insertion loss, reflection testing, and network performance validation

KU8. fiber management practices, such as slack management, connector polishing techniques, and scalability considerations

KU9. emerging tools and equipment for installation and diagnostics, including AI-enabled OTDRs and advanced OLTs

KU10. the safety protocols for handling optical fibers, tools, and testing equipment, including the use of personal protective equipment (PPE)

KU11. the documentation practices for installation processes, loss budgets, tests, and troubleshooting results

KU12. basics of Wavelength Division Multiplexing (WDM) and its role in fiber networks, including fundamental principles of coherent optics in high-speed data transmission

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** document and maintain accurate records of installations, test results, and loss budgets
- GS2.** read and interpret network layouts, technical manuals, and fiber specifications
- GS3.** communicate effectively with customers, colleagues, and stakeholders to ensure smooth installations
- GS4.** use diagnostic tools to identify and resolve network issues effectively
- GS5.** plan and prioritize installation tasks to meet deadlines while ensuring compliance with standards
- GS6.** coordinate with external vendors for sourcing and maintaining tools, components, and equipment
- GS7.** adapt to new technologies like AI-enabled diagnostics and advanced PON techniques
- GS8.** troubleshoot network issues systematically to minimize downtime and maximize efficiency
- GS9.** monitor safety compliance during all stages of the installation process

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install passive FTTH/X components (Splitters)</i>	10	20	-	7
PC1. identify and assess passive components, including PLC and FBT splitters, and their deployment environments	2	4	-	2
PC2. install wall-mount and rack-mount splitters (e.g., 1x8, 1x16, 1x32) using precision tools and advanced equipment	2	4	-	2
PC3. ensure compatibility of splitters with GPON and emerging PON technologies like XG-PON and NG-PON2	2	4	-	1
PC4. check installation sites for optimal placement to minimize signal loss, comply with building standards, and address environmental impacts	2	4	-	1
PC5. identify and configure splitters for advanced transmission technologies, including Wavelength Division Multiplexing (WDM) and Time Division Multiple Access (TDMA)	2	4	-	1
<i>Perform terminal connections (single incoming/multiple outgoing)</i>	6	10	-	4
PC6. identify and organize feeder and distribution cables/pigtails for efficient routing and minimal signal interference	2	3	-	1
PC7. use advanced connectors (e.g., SC, LC, APC) to establish secure and loss-free connections	2	3	-	1
PC8. apply fiber management techniques to ensure organized, secure, and scalable installations	1	2	-	1
PC9. perform final connector polishing to maximize optical clarity and reduce insertion loss	1	2	-	1
<i>Conduct power and insertion loss tests</i>	6	10	-	4
PC10. conduct insertion loss and reflection testing using advanced OLTS and OTDR equipment	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. measure power output at distribution ports using precision power meters	1	3	-	1
PC12. validate network performance parameters for compliance with GPON and NG-PON2 standards	1	2	-	1
PC13. analyze and calculate loss budgets, considering WDM and TDMA technologies, to ensure network design compliance	2	3	-	1
<i>Work with GPON technologies, loss budgets, and advanced transmission mechanisms</i>	8	10	-	5
PC14. install and configure passive components compatible with advanced GPON, XG-PON, and NG-PON2 networks	2	2	-	1
PC15. identify the role of broadcast technologies in GPON deployments and configure splitters accordingly	2	2	-	1
PC16. use AI-enabled diagnostic tools for real-time fault detection, insertion loss tracking, and troubleshooting	2	2	-	1
PC17. analyze the impact of components (e.g., splitters, connectors) on loss budgets and optimize designs using best practices	1	2	-	1
PC18. validate the configuration and deployment of advanced transmission mechanisms like WDM and TDMA for optimized bandwidth utilization	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4200
NOS Name	Installation of Passive FTTH/X components
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Network (Passive) Installation
NSQF Level	3
Credits	3
Version	6.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N4201: In-building FTTH/X cabling

Description

This unit is about installing fiber cables for FTTH/X deployment in buildings, including IoT, Internet of Everything (IoE), Triple-Play services, and Smart Home Network readiness. The process starts from the OSP (outside plant) fiber landing point to the ONT termination and Telecommunication Outlet (TO) in customer premises, incorporating cybersecurity measures to ensure secure installations and optimal Quality of Service (QoS)

Scope

The scope covers the following :

- Cable installation through cable trays (horizontal/vertical)
- Cable installation through conduits
- Cable installation through false ceilings
- Termination at ONT and TO
- IoT and Smart Home Network Readiness
- FTTH GPON Customer Installation
- Triple-Play Services (High Speed Internet, VoIP, IPTV)
- Cybersecurity considerations in FTTH/X installations
- Internet of Everything (IoE)

Elements and Performance Criteria

Cable installation through cable trays (horizontal/vertical)

To be competent, the user/individual on the job must be able to:

- PC1.** check the site as per the building layout plan and identify the cabling path from the outdoor fiber landing point to the ONT installation point
- PC2.** determine horizontal and vertical cable lengths, considering slack for maintenance and future upgrades
- PC3.** check load compliance of cable trays and ensure compatibility with existing services like power and data cables
- PC4.** lay fiber along tray tracks using proper pulling techniques, ensuring no damage to the cable jacket or core
- PC5.** secure fibers in the trays, maintaining proper slack and tension to avoid over-tensioning in vertical runs
- PC6.** ensure proper grounding of metallic trays in line with safety standards

Cable installation through conduits

To be competent, the user/individual on the job must be able to:

- PC7.** pull fiber through conduits using appropriate tools (e.g., fish tape, pullers) to avoid damage
- PC8.** secure excess fiber (minimum of 3 meters) at termination points for maintenance purposes
- PC9.** inspect conduit integrity to prevent electromagnetic interference or mechanical damage
- PC10.** ensure proper sealing of conduits to avoid dust, moisture, or pest intrusion

Qualification Pack

Cable installation through false ceilings

To be competent, the user/individual on the job must be able to:

- PC11.** install cables through false ceilings using the figure-8 method to prevent tangling or cable stress
- PC12.** secure cables in conduits above false ceilings to prevent dislodgement and ensure slack management
- PC13.** ensure accessibility for future maintenance by marking cable routes clearly

Termination at ONT and TO

To be competent, the user/individual on the job must be able to:

- PC14.** terminate and connectorize fiber at the ONT, ensuring signal integrity and minimal loss
- PC15.** power up and configure the ONT for operational readiness, including VLAN and IP address settings
- PC16.** conduct live fiber testing using tools like Visual Fault Locator (VFL) and power meters to confirm signal integrity
- PC17.** validate ONT connectivity to IoT devices and smart home systems, ensuring proper data throughput

IoT and Smart Home Network Readiness

To be competent, the user/individual on the job must be able to:

- PC18.** determine IoT device connectivity requirements such as bandwidth and latency, and verify network configuration
- PC19.** installation network elements or CPEs for IoT devices
- PC20.** configure ONTs to support IoT devices like smart thermostats, cameras, and voice assistants
- PC21.** test IoT device compatibility with installed FTTH networks to ensure seamless integration
- PC22.** coordinate with customers for specific IoT device setups and provide technical guidance

FTTH GPON Customer Installation

To be competent, the user/individual on the job must be able to:

- PC23.** check and prepare customer premises for installing Customer Premises Equipment (CPE), including ONTs and Wi-Fi routers
- PC24.** follow GPON installation procedures, ensuring correct splitter connections and fiber termination
- PC25.** conduct comprehensive tests for connectivity and data speeds at the customer's end using tools like OTDR and fiber testers
- PC26.** troubleshoot network issues related to CPE and ensure customer satisfaction by addressing concerns promptly
- PC27.** identify and resolve common complaints related to fiber connectivity, signal loss, and ONT/router configurations
- PC28.** use installed equipment and provide basic troubleshooting training to customers, explaining technical details in easy terms and addressing their concerns

Triple-Play Services (High Speed Internet, VoIP, IPTV)

To be competent, the user/individual on the job must be able to:

- PC29.** determine the infrastructure requirements for Triple-Play services, including separate VLANs for data, voice, and video

Qualification Pack

- PC30.** configure ONT settings to enable Triple-Play services, ensuring optimal bandwidth allocation for each
- PC31.** test High-Speed Internet, VoIP, and IPTV services for Quality of Service (QoS) parameters like latency, jitter, and packet loss
- PC32.** coordinate with service providers to address issues with VoIP call quality, IPTV buffering, or internet speeds

Cybersecurity Considerations in FTTH/X Installations

To be competent, the user/individual on the job must be able to:

- PC33.** identify potential cybersecurity vulnerabilities in FTTH installations and mitigate risks using secure installation practices
- PC34.** configure ONTs with secure settings, including password protection, encryption protocols, and firewalls
- PC35.** conduct penetration tests to identify potential security risks and validate network integrity
- PC36.** provide customers with guidelines for maintaining network security, including password updates and device firmware updates

Internet of Everything (IoE)

To be competent, the user/individual on the job must be able to:

- PC37.** identify the scope of IoE and its impact on FTTH network design and installation
- PC38.** integrate IoE-compatible devices into the FTTH network, ensuring seamless communication between devices
- PC39.** follow future trends in IoE and identify scalable network solutions for customers
- PC40.** optimize FTTH installations for emerging IoE applications, ensuring minimal latency and maximum reliability

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types and characteristics of fiber optic cables, including bend radius and tensile strength
- KU2.** the role of fiber networks in supporting cloud gaming platforms and ultra-low latency applications such as High-Frequency Trading (HFT) in financial markets
- KU3.** how fiber-optic connectivity supports Industry 4.0 applications, including automation, robotics, and real-time data monitoring in smart manufacturing
- KU4.** fusion splicing, termination techniques, and proper handling practices for fiber cables
- KU5.** IoT and IoE device types, their connectivity requirements, and network configurations
- KU6.** the role of FTTH in IoE
- KU7.** the key FTTH GPON components and their functions
- KU8.** the basics of network security, including encryption protocols, firewalls, and access control mechanisms
- KU9.** tools for fiber installation (e.g., fish tape, splicing machines, OTDR, VFL) and their proper usage
- KU10.** the techniques to measure signal loss and maintain network performance
- KU11.** Triple-Play service requirements (internet, voice, video) and their impact on network infrastructure

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- KU12.** GPON technology, including splitters, ONT configuration, and VLAN management
- KU13.** Quality of Service (QoS) parameters, such as latency, jitter, and throughput, and how to optimize them
- KU14.** cybersecurity considerations in FTTH networks, and appropriate and mitigation measures including encryption techniques, access control, and security protocols for ONT, OLT, and IoT integration
- KU15.** future trends in IoE and innovations in smart home and triple-play technologies
- KU16.** Safety, Health, and Environmental (SHE) and occupational health and safety (OHS) regulations
- KU17.** the documentation requirements for installation, testing, and cybersecurity compliance
- KU18.** AI-driven network management and automation tools for monitoring fiber performance and diagnosing faults remotely

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret building layouts, network diagrams, and IoT/IoE device manuals
- GS2.** communicate effectively with customers, colleagues, and supervisors for smooth installations
- GS3.** maintain accurate documentation of installations, testing, and cybersecurity measures
- GS4.** identify and mitigate risks related to fiber installation, Triple-Play services, and network security
- GS5.** use digital tools for configuring ONTs, testing networks, and troubleshooting
- GS6.** coordinate with third-party vendors for equipment, repairs, and cybersecurity audits
- GS7.** analyze test results to identify and resolve network issues efficiently
- GS8.** adapt to new technologies such as AI-enabled diagnostics and IoE applications
- GS9.** prioritize tasks to meet deadlines and maintain customer satisfaction
- GS10.** follow ethical practices while handling customer data and maintaining network security

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cable installation through cable trays (horizontal/vertical)</i>	6	10	-	3
PC1. check the site as per the building layout plan and identify the cabling path from the outdoor fiber landing point to the ONT installation point	1	2	-	0.5
PC2. determine horizontal and vertical cable lengths, considering slack for maintenance and future upgrades	1	2	-	0.5
PC3. check load compliance of cable trays and ensure compatibility with existing services like power and data cables	1	2	-	0.5
PC4. lay fiber along tray tracks using proper pulling techniques, ensuring no damage to the cable jacket or core	1	2	-	0.5
PC5. secure fibers in the trays, maintaining proper slack and tension to avoid over-tensioning in vertical runs	1	1	-	0.5
PC6. ensure proper grounding of metallic trays in line with safety standards	1	1	-	0.5
<i>Cable installation through conduits</i>	4	4	-	2
PC7. pull fiber through conduits using appropriate tools (e.g., fish tape, pullers) to avoid damage	1	1	-	0.5
PC8. secure excess fiber (minimum of 3 meters) at termination points for maintenance purposes	1	1	-	0.5
PC9. inspect conduit integrity to prevent electromagnetic interference or mechanical damage	1	1	-	0.5
PC10. ensure proper sealing of conduits to avoid dust, moisture, or pest intrusion	1	1	-	0.5
<i>Cable installation through false ceilings</i>	2	4	-	2
PC11. install cables through false ceilings using the figure-8 method to prevent tangling or cable stress	1	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. secure cables in conduits above false ceilings to prevent dislodgement and ensure slack management	0.5	2	-	0.5
PC13. ensure accessibility for future maintenance by marking cable routes clearly	0.5	1	-	0.5
<i>Termination at ONT and TO</i>	3	4	-	2
PC14. terminate and connectorize fiber at the ONT, ensuring signal integrity and minimal loss	1	1	-	0.5
PC15. power up and configure the ONT for operational readiness, including VLAN and IP address settings	1	1	-	0.5
PC16. conduct live fiber testing using tools like Visual Fault Locator (VFL) and power meters to confirm signal integrity	0.5	1	-	0.5
PC17. validate ONT connectivity to IoT devices and smart home systems, ensuring proper data throughput	0.5	1	-	0.5
<i>IoT and Smart Home Network Readiness</i>	3	5	-	2
PC18. determine IoT device connectivity requirements such as bandwidth and latency, and verify network configuration	0.5	1	-	0.5
PC19. installation network elements or CPEs for IoT devices	0.5	1	-	0.5
PC20. configure ONTs to support IoT devices like smart thermostats, cameras, and voice assistants	1	1	-	0.5
PC21. test IoT device compatibility with installed FTTH networks to ensure seamless integration	0.5	1	-	-
PC22. coordinate with customers for specific IoT device setups and provide technical guidance	0.5	1	-	0.5
<i>FTTH GPON Customer Installation</i>	4	6	-	2
PC23. check and prepare customer premises for installing Customer Premises Equipment (CPE), including ONTs and Wi-Fi routers	1	1	-	0.5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. follow GPON installation procedures, ensuring correct splitter connections and fiber termination	1	1	-	0.5
PC25. conduct comprehensive tests for connectivity and data speeds at the customer's end using tools like OTDR and fiber testers	0.5	1	-	0.5
PC26. troubleshoot network issues related to CPE and ensure customer satisfaction by addressing concerns promptly	0.5	1	-	-
PC27. identify and resolve common complaints related to fiber connectivity, signal loss, and ONT/router configurations	0.5	1	-	0.5
PC28. use installed equipment and provide basic troubleshooting training to customers, explaining technical details in easy terms and addressing their concerns	0.5	1	-	-
<i>Triple-Play Services (High Speed Internet, VoIP, IPTV)</i>	3	5	-	2
PC29. determine the infrastructure requirements for Triple-Play services, including separate VLANs for data, voice, and video	1	2	-	0.5
PC30. configure ONT settings to enable Triple-Play services, ensuring optimal bandwidth allocation for each	1	1	-	0.5
PC31. test High-Speed Internet, VoIP, and IPTV services for Quality of Service (QoS) parameters like latency, jitter, and packet loss	0.5	1	-	0.5
PC32. coordinate with service providers to address issues with VoIP call quality, IPTV buffering, or internet speeds	0.5	1	-	0.5
<i>Cybersecurity Considerations in FTTH/X Installations</i>	3	6	-	2
PC33. identify potential cybersecurity vulnerabilities in FTTH installations and mitigate risks using secure installation practices	1	1	-	0.5
PC34. configure ONTs with secure settings, including password protection, encryption protocols, and firewalls	1	1	-	0.5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. conduct penetration tests to identify potential security risks and validate network integrity	0.5	2	-	0.5
PC36. provide customers with guidelines for maintaining network security, including password updates and device firmware updates	0.5	2	-	0.5
<i>Internet of Everything (IoE)</i>	2	6	-	3
PC37. identify the scope of IoE and its impact on FTTH network design and installation	0.5	2	-	1
PC38. integrate IoE-compatible devices into the FTTH network, ensuring seamless communication between devices	0.5	2	-	1
PC39. follow future trends in IoE and identify scalable network solutions for customers	0.5	1	-	0.5
PC40. optimize FTTH installations for emerging IoE applications, ensuring minimal latency and maximum reliability	0.5	1	-	0.5
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4201
NOS Name	In-building FTTH/X cabling
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Network (Passive) Installation
NSQF Level	3
Credits	1
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N4131: Work Safety Practices with Fiber Optics

Description

This unit is about work safety practices whilst working with fiber optics, with specific updates to include safety protocols for Fiber-to-the-Home (FTTH/X) installations

Scope

The scope covers the following :

- Safety practices for working with fiber optics during FTTH/X installations
- Industry standards and guidelines

Elements and Performance Criteria

Safety practices for working with fiber optics during FTTH/X installations

To be competent, the user/individual on the job must be able to:

- PC1.** follow appropriate eye-safety measures whilst working with laser-emitting devices, including Optical Network Terminals (ONTs) and splicing equipment
- PC2.** safely handle bare fiber, broken ends, and scraps, ensuring proper disposal to prevent accidental injury
- PC3.** follow the recommended fire safety practices, especially when using high-voltage arc fusion splicers and other heating tools
- PC4.** adhere to electrical safety norms when working with fiber cables installed alongside electrical cables and active power sources
- PC5.** follow laser safety rules, including proper handling of class 1m and higher laser devices used in FTTH networks
- PC6.** use and maintain appropriate safety gear, such as gloves, boots, and protective eyewear, in accordance with worksite requirements
- PC7.** follow industry standards and guidelines (e.g., ANSI Z136.2 for laser safety, ISO standards for fiber optic installations)
- PC8.** maintain health and safety records accurately, including incident reports and compliance checklists
- PC9.** identify and mitigate hazards associated with installation sites, such as confined spaces, sharp edges, and high temperatures
- PC10.** examine and safely handle pre-terminated fiber assemblies and connectors to prevent contamination or damage
- PC11.** apply safe cable routing techniques to avoid damage to existing infrastructure, such as gas pipelines, electrical cables, and water pipelines

Industry standards and guidelines

To be competent, the user/individual on the job must be able to:

- PC12.** follow national and international guidelines for fiber optic installations, such as TIA/EIA-568, ITU-T standards, and BICSI guidelines

Qualification Pack

- PC13.** comply with regulatory and environmental norms related to FTTH/X deployment, including e-waste management and proper waste disposal practices
- PC14.** identify and report non-compliance issues or deviations from industry standards to the supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** construction of fiber optics and methods for protecting fibers from environmental damage
- KU2.** the latest PPE required for fiber optic installations such as safety glasses, cut-resistant gloves, protective clothing, safety footwear, hard hats, hearing protection, high-visibility vests, and respiratory protection
- KU3.** the benefits of using PPE such as ensuring safety, preventing injuries, reducing workplace accidents, and ensuring compliance with occupational safety regulations
- KU4.** the safety features, limitations, and maintenance protocols for protective equipment and gear
- KU5.** laser safety guidelines, including risk levels of various laser classes used in fiber optics
- KU6.** the hazards associated with optical fibers, such as microshards and laser exposure, and safe disposal practices for fiber scraps and hazardous materials
- KU7.** layout of associated services in the work area (gas pipelines, electrical cables, sewage lines, water pipelines, etc.) to avoid consequential damage during FTTH installation
- KU8.** key industry standards and guidelines for fiber optic installations (e.g., ANSI/TIA, ITU-T, ISO)
- KU9.** compliance requirements for environmental regulations and e-waste management in FTTH installations
- KU10.** contact details of government agencies and regulatory bodies responsible for safety, health, and security
- KU11.** importance of maintaining accurate and timely documentation to ensure compliance with industry and organizational standards
- KU12.** procedures for conducting periodic audits to ensure adherence to safety standards
- KU13.** manufacturer-supplied MSDS for safe handling of fiber optic materials and chemicals
- KU14.** legislative requirements and organizational procedures for health, safety, and security, and the roles and responsibilities of individuals
- KU15.** different types of health and safety hazards in FTTH/X installation sites
- KU16.** limits of personal responsibility for dealing with hazards and escalating critical issues
- KU17.** different emergency situations, including handling accidental fiber cuts, exposure to high voltage, and adhering to emergency procedures
- KU18.** the importance of maintaining high standards of health, safety, and security
- KU19.** the implications of non-compliance with health, safety, and security protocols on individuals and organizations

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** read and interpret necessary documents, including safety manuals, standards, and compliance guidelines
- GS2.** understand and follow operational health and safety instructions, memos, and reports
- GS3.** read and comprehend technical specifications related to FTTH/X installations
- GS4.** communicate effectively with colleagues, supervisors, and stakeholders to report hazards, seek guidance, and ensure smooth operations
- GS5.** liaise and coordinate with third-party vendors and contractors to ensure adherence to safety protocols
- GS6.** document safety incidents, near misses, and compliance checklists clearly and accurately
- GS7.** adapt to new technologies and safety equipment introduced in FTTH/X installations
- GS8.** manage time effectively to prioritize safety-related tasks without compromising installation timelines
- GS9.** identify and report unsafe conditions or practices to the appropriate authority
- GS10.** use digital tools and software for safety monitoring, documentation, and compliance tracking

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Safety practices for working with fiber optics during FTTH/X installations</i>	20	30	-	15
PC1. follow appropriate eye-safety measures whilst working with laser-emitting devices, including Optical Network Terminals (ONTs) and splicing equipment	2	3	-	2
PC2. safely handle bare fiber, broken ends, and scraps, ensuring proper disposal to prevent accidental injury	2	3	-	2
PC3. follow the recommended fire safety practices, especially when using high-voltage arc fusion splicers and other heating tools	2	3	-	1
PC4. adhere to electrical safety norms when working with fiber cables installed alongside electrical cables and active power sources	2	3	-	1
PC5. follow laser safety rules, including proper handling of class 1m and higher laser devices used in FTTH networks	2	3	-	2
PC6. use and maintain appropriate safety gear, such as gloves, boots, and protective eyewear, in accordance with worksite requirements	2	3	-	2
PC7. follow industry standards and guidelines (e.g., ANSI Z136.2 for laser safety, ISO standards for fiber optic installations)	2	3	-	1
PC8. maintain health and safety records accurately, including incident reports and compliance checklists	2	3	-	1
PC9. identify and mitigate hazards associated with installation sites, such as confined spaces, sharp edges, and high temperatures	2	2	-	1
PC10. examine and safely handle pre-terminated fiber assemblies and connectors to prevent contamination or damage	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. apply safe cable routing techniques to avoid damage to existing infrastructure, such as gas pipelines, electrical cables, and water pipelines	1	2	-	1
<i>Industry standards and guidelines</i>	10	20	-	5
PC12. follow national and international guidelines for fiber optic installations, such as TIA/EIA-568, ITU-T standards, and BICSI guidelines	4	8	-	2
PC13. comply with regulatory and environmental norms related to FTTH/X deployment, including e-waste management and proper waste disposal practices	3	6	-	2
PC14. identify and report non-compliance issues or deviations from industry standards to the supervisor	3	6	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N4131
NOS Name	Work Safety Practices with Fiber Optics
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Operations and Maintenance - Passive Infrastructure
NSQF Level	3
Credits	1
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N9111: Follow sustainability practices in telecom cabling operations

Description

This unit covers the application of sustainable and environmentally responsible practices in telecom cabling operations. It includes identifying recyclable and hazardous materials, implementing effective waste management strategies, optimizing resource usage, and ensuring compliance with environmental regulations.

Scope

The scope covers the following :

- Identify recyclable, reusable, and hazardous materials
- Follow waste management, recycling, and disposal protocols
- Optimize material and energy usage in cabling work
- Comply with environmental and regulatory standards

Elements and Performance Criteria

Identify recyclable, reusable, and hazardous materials

To be competent, the user/individual on the job must be able to:

- PC1.** identify different types of materials used in cabling operations, such as optical fiber, connectors, protective ducts, and joint enclosures
- PC2.** segregate materials into recyclable (e.g., plastic ducts, fiber offcuts, packaging), reusable (e.g., spare connectors, splitters), and hazardous (e.g., lead-based cables, old batteries, chemical adhesives) categories
- PC3.** follow workplace procedures for storing recyclable materials separately from hazardous and general waste
- PC4.** label and store hazardous waste safely to prevent contamination or accidents during handling and disposal
- PC5.** ensure all recyclable and hazardous waste is recorded in logs for traceability and compliance

Follow waste management, recycling, and disposal protocols

To be competent, the user/individual on the job must be able to:

- PC6.** adhere to applicable SOPs for safe handling and disposal of non-recyclable and hazardous materials, such as fiber shards, cable sheaths, and chemical adhesives
- PC7.** coordinate safe disposal of waste materials at designated collection points or through approved vendors, as per organizational policies and environmental guidelines
- PC8.** reduce waste by minimizing excess material use and reusing components wherever possible
- PC9.** maintain clean and organized work sites to prevent environmental contamination and promote workplace safety
- PC10.** check for and report any issues related to improper waste disposal, unauthorized dumping, or environmental hazards to the relevant authority

Optimize material and energy usage in cabling work

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC11.** plan cabling work to minimize material wastage by accurately measuring fiber optic lengths, ducts, and accessories before cutting or installing
- PC12.** use energy-efficient tools and machinery where possible, ensuring minimal power consumption without compromising work efficiency
- PC13.** ensure proper maintenance of cabling tools and equipment to reduce unnecessary repairs and replacements that contribute to waste
- PC14.** optimize the use of trenching and ducting by following efficient layout designs to minimize excavation and reduce material consumption
- PC15.** use and promote eco-friendly materials (such as low-impact protective coatings and biodegradable packaging) wherever feasible

Comply with environmental and regulatory standards

To be competent, the user/individual on the job must be able to:

- PC16.** follow national and local environmental laws, regulations, and workplace policies related to telecom cabling operations
- PC17.** maintain proper documentation of sustainability-related activities, including logs of disposed and recycled materials, to meet regulatory and audit requirements
- PC18.** conduct periodic self-audits to verify compliance with environmental policies
- PC19.** educate team members on best practices for sustainability, including waste segregation, material reuse, and responsible energy consumption
- PC20.** report any violations of environmental policies, hazardous material spills, or unsafe disposal practices to the designated supervisor or regulatory body

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies on sustainability, waste reduction, and material reuse in telecom infrastructure projects
- KU2.** the procedures for recycling, hazardous waste handling, and safe disposal of telecom-related materials
- KU3.** the importance of sustainability in long-term infrastructure planning, including the impact of telecom waste on the environment
- KU4.** the classification of materials used in optical fiber cabling, including recyclable, reusable, and hazardous components
- KU5.** standard waste management procedures for telecom operations, including segregation, labeling, and disposal methods
- KU6.** the methods to reduce material wastage, such as accurate measurements, careful handling of fiber optic cables, and optimized trenching techniques
- KU7.** the environmental hazards associated with improper disposal of optical fibers, batteries, and chemical adhesives
- KU8.** the regulations and compliance requirements for hazardous material disposal under national and international environmental laws
- KU9.** energy-efficient work practices, including low-power tools, optimized route planning, and reduced excavation techniques

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- KU10.** the importance of proper record-keeping for disposal and recycling to ensure compliance and accountability

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively sustainability practices and compliance requirements to team members and supervisors
- GS2.** identify and implement solutions for reducing material wastage and improving recycling efforts
- GS3.** select the best disposal or recycling methods based on regulatory compliance and workplace policies
- GS4.** determine the environmental impact of cabling operations and suggest improvements
- GS5.** work effectively with waste management teams, regulatory bodies, and supervisors to ensure compliance
- GS6.** accurately document materials used, disposed of, or recycled for audit and compliance purposes
- GS7.** prioritize sustainability initiatives while meeting project deadlines
- GS8.** incorporate new sustainable materials and methods into existing work processes

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify recyclable, reusable, and hazardous materials</i>	8	15	-	5
PC1. identify different types of materials used in cabling operations, such as optical fiber, connectors, protective ducts, and joint enclosures	2	3	-	1
PC2. segregate materials into recyclable (e.g., plastic ducts, fiber offcuts, packaging), reusable (e.g., spare connectors, splitters), and hazardous (e.g., lead-based cables, old batteries, chemical adhesives) categories	2	3	-	1
PC3. follow workplace procedures for storing recyclable materials separately from hazardous and general waste	2	3	-	1
PC4. label and store hazardous waste safely to prevent contamination or accidents during handling and disposal	1	3	-	1
PC5. ensure all recyclable and hazardous waste is recorded in logs for traceability and compliance	1	3	-	1
<i>Follow waste management, recycling, and disposal protocols</i>	6	15	-	5
PC6. adhere to applicable SOPs for safe handling and disposal of non-recyclable and hazardous materials, such as fiber shards, cable sheaths, and chemical adhesives	2	3	-	1
PC7. coordinate safe disposal of waste materials at designated collection points or through approved vendors, as per organizational policies and environmental guidelines	1	3	-	1
PC8. reduce waste by minimizing excess material use and reusing components wherever possible	1	3	-	1
PC9. maintain clean and organized work sites to prevent environmental contamination and promote workplace safety	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. check for and report any issues related to improper waste disposal, unauthorized dumping, or environmental hazards to the relevant authority	1	3	-	1
<i>Optimize material and energy usage in cabling work</i>	8	10	-	5
PC11. plan cabling work to minimize material wastage by accurately measuring fiber optic lengths, ducts, and accessories before cutting or installing	2	2	-	1
PC12. use energy-efficient tools and machinery where possible, ensuring minimal power consumption without compromising work efficiency	2	2	-	1
PC13. ensure proper maintenance of cabling tools and equipment to reduce unnecessary repairs and replacements that contribute to waste	2	2	-	1
PC14. optimize the use of trenching and ducting by following efficient layout designs to minimize excavation and reduce material consumption	1	2	-	1
PC15. use and promote eco-friendly materials (such as low-impact protective coatings and biodegradable packaging) wherever feasible	1	2	-	1
<i>Comply with environmental and regulatory standards</i>	8	10	-	5
PC16. follow national and local environmental laws, regulations, and workplace policies related to telecom cabling operations	2	2	-	1
PC17. maintain proper documentation of sustainability-related activities, including logs of disposed and recycled materials, to meet regulatory and audit requirements	2	2	-	1
PC18. conduct periodic self-audits to verify compliance with environmental policies	2	2	-	1
PC19. educate team members on best practices for sustainability, including waste segregation, material reuse, and responsible energy consumption	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. report any violations of environmental policies, hazardous material spills, or unsafe disposal practices to the designated supervisor or regulatory body	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9111
NOS Name	Follow sustainability practices in telecom cabling operations
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N4143.Install Fiber-to-the-Home (FTTH/X) and Fiber-to-Anywhere (FTTx) Cables	30	50	-	20	100	15
TEL/N6400.Splice Optical Fiber	30	50	-	20	100	15
TEL/N4200.Installation of Passive FTTH/X components	30	50	-	20	100	15
TEL/N4201.In-building FTTH/X cabling	30	50	-	20	100	15
TEL/N4131.Work Safety Practices with Fiber Optics	30	50	0	20	100	15
TEL/N9111.Follow sustainability practices in telecom cabling operations	30	50	0	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	200	330	0	120	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
TSSC	Telecom Sector Skill Council
TRAI	Telecom Regulatory Authority of India
FTTx	Fiber to the x
FTTN	Fiber to the Node
FTTC	Fiber to the Curb
FTTB	Fiber to the Building
PON	Passive Optical Network
GPON	Gigabit Passive Optical Network
OTDR	Optical Time Domain Reflectometer
OSA	Optical Spectrum Analyzer
CD	Chromatic Dispersion
PMD	Polarization Mode Dispersion
OFIs	Optical Fiber Installations

Qualification Pack

NOC	Network Operations Center
OSP	Outside Plant
WDM	Wavelength Division Multiplexing
TDMA	Time Division Multiple Access
ONT	Optical Network Terminal
VFL	Visual Fault Locator
CPE	Customer Premises Equipment

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.